

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 6+ years building scalable backend services and React frontends on the MERN stack (MongoDB, Express.js, React, Node.js) with TypeScript. Led the zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, reporting directly to the CTO, and built real-time data pipelines that kept live systems consistent throughout. Experienced with microservices architecture (Apache Kafka, Docker, Kubernetes), third-party integrations, and shipping products for African markets — including USSD services for users without smartphones. Now applying that reliability-first systems thinking to financial services used by millions.

CORE COMPETENCIES

Node.js & TypeScript

MERN Stack (MongoDB, Express, React, Node)

Microservices Architecture

Docker & Kubernetes

MySQL & MongoDB

Third-Party Integrations

AWS & GCP

Agile Delivery

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing
- Integrated software components and third-party services across the stack; migrated a Webflow website from a Parse backend to Firebase
- Implemented Express Shipping, generating an additional \$40 revenue per order; built a 3D visualization feature (video overlay + ffmpeg) that increased buyer conversion
- Saved the company \$5,000/month by migrating all services off AWS; moved file storage from Amazon S3 to GCS via a Python script

MTailor (Contract)

Mar 2022 – Jul 2022

Backend Migration Contractor — US Remote

- Built a migration script reading from MongoDB and writing to Firestore after complex processing logic
- Wrote technical documentation for fellow engineers on the new Firebase SDKs (Node.js and Python); trained the Ops team on dashboards and backup procedures

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Led optimization of a PWA delivering real-time stock market prices; built jobs querying **2M+ Firestore records** to keep prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js; added responsive React UI pages with Tailwind

Mind2matter

Jan 2022 – Jul 2022

Software Engineer (Full-Stack) — US Remote

- Built backends for DeFi applications using Web3 and Node.js; delivered React UIs from Figma designs under tight timelines for demanding clients

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead — Uganda

- Led a team of 4 developers building legal tech systems for NGOs in Uganda; built case management for FIDA Uganda and a paralegal database for LASPNET
- Architected a microservices backend using **Apache Kafka, Docker, and Kubernetes** for FIDA Uganda's case management app
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC — production software for African-market constraints
- Developed a cross-platform mobile app with React Native and Expo (iOS & Android) with Firebase push notifications

PROJECTS

Real-time MongoDB ↔ Firestore Sync Pipeline

Live two-way sync pipeline supporting parallel production traffic during a 20+ app migration — idempotent message processing over Google Pub/Sub.

Node.js, MongoDB, Firestore, Google Pub/Sub

Healthcare WhatsApp Chatbot

Integrated OpenAI and Google Vision for prescription OCR via the WhatsApp Business API — third-party integration across three vendor APIs.

Node.js, WhatsApp Business API, OpenAI, Google Vision

Marketplace Recommendation Engine

Collaborative and content-based filtering engine in MongoDB for service provider matching.

Node.js, MongoDB

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, MongoDB, SQL (MySQL), Express.js, Firebase, GCP, Python, Git, Linux

Intermediate: Go, AWS, Rust, React Native **Infrastructure:** Docker, Kubernetes, Apache Kafka, NATS Streaming, gRPC

Languages: English, Kiswahili